CHAPTER 17

Acids and bases

Acids and bases have an important and diverse role. They are found in homes and are used extensively in industry and agriculture. Acids and bases are also the reactants and products of many chemical reactions that take place in environmental and biological systems.

In this chapter, you will study a theory that explains the characteristic reactions of acids and bases. You will learn to represent the common reactions of acids and bases using ionic equations and investigate the distinction between strong and weak acids, and concentrated and dilute acids. Finally, you will learn how to calculate the pH of an acid and determine the concentration of unknown acids and bases using stoichiometry.

Science understanding

- indicator colour and the pH scale are used to classify aqueous solutions as acidic, basic or neutral
- pH is used as a measure of the acidity of solutions and is dependent on the concentration of hydrogen ions in the solution
- the Arrhenius model can be used to explain the behaviour of strong and weak acids and bases in aqueous solutions
- patterns of the reactions of acids and bases, including reactions of acids with bases, metals and carbonates and the reactions of bases with acids and ammonium salts, allow products and observations to be predicted from reactants; ionic equations represent the reacting species and products in these reactions
- the presence of specific ions in solutions can be identified by observing the colour of the solution, flame tests and observing various chemical reactions, including precipitation and acid-base reactions
- the mole concept can be used to calculate the mass of solute, and solution concentrations and volumes involved in a chemical reaction

17.1 Properties of acids and bases

Acids and **bases** make up some of the household products in your kitchen and laundry (Figure 17.1.1). In this section, you will be introduced to a theory that explains the chemical properties of acids and bases, helping you to explain their usefulness within the home and industry. You will also look at how the **acidity** of a solution can be measured, therefore defining a solution as a strong or weak acid.

FIGURE 17.1.1 Common household products that contain acids, bases or salts

ACIDS AND BASES

Table 17.1.1 gives the names, chemical formulae and uses of some common acids.

TABLE 17.1.1 Common acids and their everyday uses

Name	Formula	Uses
hydrochloric acid	HCl	present in stomach acid to help break down proteins
sulfuric acid	H_2SO_4	used in car batteries and in the manufacture of fertilisers and detergents
nitric acid	HNO_3	used in the manufacture of fertilisers, dyes and explosives
ethanoic acid (acetic acid)	CH_3COOH	found in vinegar; used as a preservative
carbonic acid	H_2CO_3	found in carbonated soft drinks
citric acid	$H_8C_6O_7$	found in the juice of citrus fruits, such as lemons

Many cleaning agents used in the home, such as washing powders and oven cleaners, contain bases. Solutions of ammonia are used as floor cleaners, and sodium hydroxide is the major active ingredient in oven cleaner. Bases are effective cleaners because they react with fats or oils to produce water-soluble soaps. A soluble base is referred to as an **alkali**.

Soluble bases are called alkalis.

Table 17.1.2 gives the names, chemical formulae and uses of some common bases.

TABLE 17.1.2 Common bases and their uses

Name	Formula	Uses
sodium hydroxide	$NaOH$	used in drain and oven cleaners, and soap making
ammonia	NH_3	used in household cleaners, fertilisers and explosives
calcium hydroxide	$Ca(OH)_2$	found in cement and mortar; used in garden lime to adjust soil pH

PROPERTIES OF ACIDS AND BASES

All acids have some properties in common. Bases also have common properties. The properties of acids and bases are summarised in Table 17.1.3.

TABLE 17.1.3 Properties of acids and bases

Properties of acids	Properties of bases
turn litmus indicator red	turn litmus indicator blue
tend to be corrosive	are caustic and feel slippery
taste sour	taste bitter
react with bases	react with acids
solutions have a pH of less than 7	solutions have a pH of greater than 7
solutions conduct an electric current	solutions conduct an electric current

INDICATORS

One of the characteristic properties of acids and bases is their ability to change the colour of certain plant extracts. Litmus is a purple dye obtained from lichen. In the presence of acids, litmus turns red. The colouring of rose petals, blackberries and red cabbage is also altered by acids and bases. Such plant extracts are called **indicators**.

Universal indicator and the pH scale

Universal indicator (Figure 17.1.2) is widely used to estimate the pH of a solution. It is a mixture of several indicators and changes through a range of colours, from red through yellow, green and blue, to violet, depending on the pH of the solution. If a more accurate measurement of pH is needed, a pH meter can be used instead of universal indicator.

FIGURE 17.1.2 Universal indicator pH scale. When universal indicator is added to a solution, it changes colour depending on the solution's pH. The tubes contain solutions of pH 0 to 14 from left to right. The green tube (centre) is neutral, pH 7.

17.1 Review

SUMMARY

- Acids have a range of uses and are often found in food and drink products.
- Bases are often used in cleaning products.
- A soluble base is called an alkali.
- Indicators are chemicals that change colour, dependent on the acidity of a solution.
- The level of acidity or alkalinity of a solution can be measured using the pH scale.
- Colours shown by universal indicator can be used to determine the approximate pH of solutions.
- Acidic solutions have a pH of less than 7 and alkaline solutions have a pH of greater than 7.

KEY QUESTIONS

1 Which one of the following acids is not usually found in food or drinks?

A hydrochloric acid
B ethanoic acid
C carbonic acid
D citric acid

2 Which one of the following is not a property of an alkaline solution?

A pH of greater than 7
B taste sour
C conduct an electric current
D turn litmus blue

3 Explain why universal indicator tells us more about an acidic solution than litmus indicator.

4 Copper(II) oxide (CuO) is a base. A few grams of copper(II) oxide were added to a beaker of distilled water and the mixture stirred. When universal indicator solution was added to the water in the beaker, the indicator was green, showing that the water was still neutral with a pH of 7. Explain this observation.

17.2 The Arrhenius model of acids and bases

There have been many attempts to define acids and bases. At first, acids and bases were just defined in terms of their observed properties such as taste, effect on an indicator and reactions with other substances. In 1887, the Swedish scientist Svante Arrhenius passed an electric current through various solutions, including acids and bases. As a result of his observations, he developed a theory that explained some of the common properties of acids and bases.

THE ARRHENIUS MODEL OF ACIDS

In the **Arrhenius model**, an acid is defined as a substance that is ionised in water to produce hydrogen ions. For example, when hydrogen chloride gas is bubbled through water, the HCl molecules break apart (a process called dissociation) and form ions (a process called ionisation) to form hydrochloric acid, which consists of hydrogen ions and chloride ions in the aqueous solution:

$$HCl(g) \rightarrow H^+(aq) + Cl^-(aq)$$

In water the H^+ ion exists in water as a hydronium ion (H_3O^+). The equation above may also be written as:

$$HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$$

In this chapter the hydrogen ion in water will be represented as $H^+(aq)$.

The Arrhenius model is useful because it explains the similarity in the reactions of different acids. The hydrogen ions that are present in acid solutions account for the common properties of acids. For example, as you will see later in this chapter, the hydrogen ions present in all acidic solutions form hydrogen gas when the acid reacts with metals such as magnesium:

$$2H^+(aq) + Mg(s) \rightarrow H_2(g) + Mg^{2+}(aq)$$

Ionisation is:

- the removal of one or more electrons from an atom or ion;
- the reaction of a molecular substance with a solvent to form ions in solution.

Polyprotic acids

Some acids can react with water to form more than one hydrogen ion per molecule. The molecules of these acids have more than one proton that can be ionised and are said to be **polyprotic acids**.

The number of hydrogen ions an acid can donate depends on the structure of the acid molecule. It is difficult to judge how many hydrogen ions can be produced by an acid by looking at its formula alone.

Ethanoic (acetic) acid, CH_3COOH, contains four hydrogen atoms, yet each molecule can only form one hydrogen ion and it is therefore a **monoprotic acid**. Only the hydrogen that is part of the highly polar O–H bond is ionised in water. In general, hydrogen atoms bonded to a highly electronegative atom, thus forming polar bonds, are ionised in solution.

Diprotic acids

Diprotic acids, such as sulfuric acid (H_2SO_4) and carbonic acid (H_2CO_3), can donate two protons.

Overall, according to the Arrhenius model, the ionisation can be represented as shown here:

$$H_2SO_4(l) \rightarrow SO_4^{2-}(aq) + 2H^+(aq)$$

In reality, a diprotic acid such as sulfuric acid ionises in two stages. You will learn more about this process in Year 12.

Triprotic acids

Triprotic acids can donate three protons. These include phosphoric acid (H_3PO_4) and boric acid (H_3BO_3). Each molecule can provide three hydrogen ions when dissolved in water as is shown here:

$$H_3PO_4(aq) \rightleftharpoons PO_4^{3-}(aq) + 3H^+(aq)$$

Note that in this equation, a reversible arrow ($\rightleftharpoons$) is used. This indicates that phosphoric acid is a weak acid. The differences between weak and strong acids are discussed in the next section.

The structure and bonding exhibited by ethanoic acid, sulfuric acid and phosphoric are shown in Figure 17.2.1.

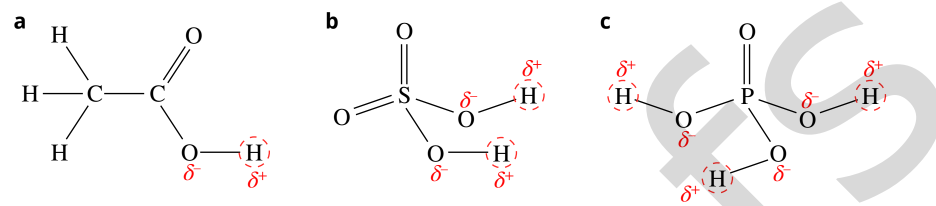


FIGURE 17.2.1 The bonds between a hydrogen atom attached to an electronegative oxygen atom are broken when these molecules are ionised in water. (a) ethanoic acid, (b) sulfuric acid, (c) phosphoric acid. The hydrogen atoms that become ions are circled.

THE ARRHENIUS MODEL OF BASES

Arrhenius defined a base as a substance that dissociates in water to form **hydroxide ions**. Sodium hydroxide is an ionic compound, containing sodium ions and hydroxide ions. When it is added to water it dissociates:

$$NaOH(s) \rightarrow Na^+(aq) + OH^-(aq)$$

Some bases dissociate to form more than one hydroxide ion:

$$Ca(OH)_2(s) \rightarrow Ca^{2+}(aq) + 2OH^-(aq)$$

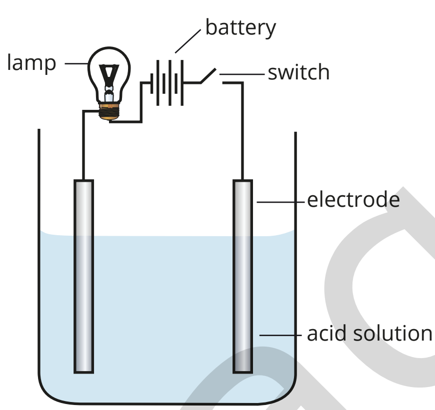
The presence of the hydroxide ions in solutions of bases accounts for the common properties of bases. For example, the hydroxide ion reacts with the litmus indicator to form a blue-coloured compound.

Ionic substances dissolve by dissociation. Ion–dipole bonds are formed between the ions and water molecules.

ACIDS AND BASES AS ELECTROLYTES

Hydrochloric acid is formed when hydrogen chloride ionises in water to form a solution containing hydrogen ions and chloride ions. When a solution of hydrochloric acid is tested with the conductivity apparatus shown in Figure 17.2.2, the light globe will glow. The chloride ions are attracted to the positive electrode of the apparatus and the hydrogen ions to the negative electrode. The flow of current in the solution is a result of the movement of these ions.

Because sodium hydroxide dissociates into sodium and hydroxide ions in aqueous solution, a sodium hydroxide solution will also conduct an electric current.



If the solution conducts an electric current, the lamp glows.

FIGURE 17.2.2 Apparatus used to test the conductivity of acid solutions

According to the Arrhenius model, acids ionise in water to produce hydrogen ions (H^+), whereas bases ionise in water to produce hydroxide ions (OH^-). The production of these charged particles in solution allows acids and bases to conduct electricity.

Neutralisation reactions

According to the Arrhenius model, when acids and bases react together, the hydrogen and hydroxide ions combine to form water according to the equation:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

Arrhenius called this a **neutralisation reaction**. You will look at neutralisation reactions in more detail in the Section 17.3.

Limitations of the Arrhenius model of acids and bases

The Arrhenius model explains the properties of acids in terms of the formation of hydrogen ions by ionisation of the acid molecule. However, the model does have some limitations.

The model does not explain why some substances that do not contain hydrogen form acidic solutions when mixed with water. For example, acidic solutions are formed when carbon dioxide (CO_2) or sulfur dioxide (SO_2), when dissolved in water. Also, the model does not explain why some substances such as ammonia (NH_3) or sodium hydrogen carbonate ($NaHCO_3$) form basic solutions when mixed with water, even though they do not contain hydroxide ions.

The Arrhenius model is also restricted to acids and bases that dissolve in water. The model does not explain acid–base behaviour in non-aqueous solutions.

An improved model for acid-base behaviour is the Brønsted–Lowry theory, that describes an acid as a substance that can donate a hydrogen ion (proton) to a base, which accepts the proton. You will learn more about the Brønsted–Lowry theory in Year 12.

ACID AND BASE STRENGTH

The acidic solution in the left-hand beaker shown in Figure 17.2.3 is more acidic, as can be seen from the more vigorous reaction with the zinc metal. If the solutions in the two beakers are of the same concentration, both contain zinc and the temperature of the acidic solutions are the same, the acid in the left-hand beaker must be a stronger acid than the acid in the beaker on the right.

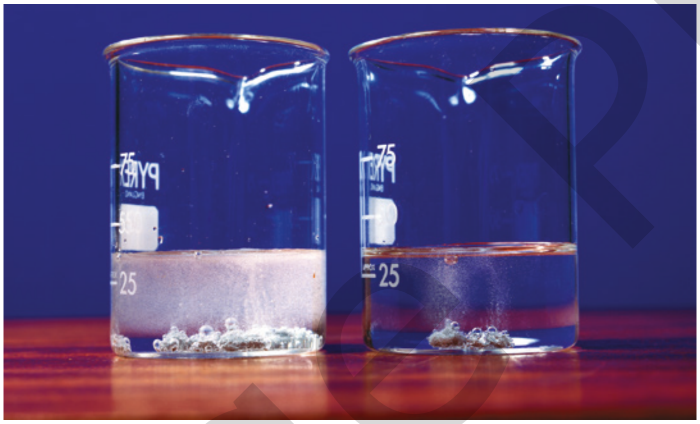


FIGURE 17.2.3 Zinc reacts more vigorously with a strong acid (left) than with a weak acid (right). The acid solutions are of equal concentration and volume.

Experiments show that different acid solutions of the same concentration do not have the same concentrations of hydrogen ions.

In the Arrhenius model, an acid is defined as a substance that is ionised in water to produce hydrogen ions. Some acids can be ionised more readily than others which makes them stronger acids. Likewise, some bases can be ionised more readily than others.

Table 17.2.1 gives the names and chemical formulae of some strong and weak acids and bases.

TABLE 17.2.1 Examples of common strong and weak acids and bases

Strong acids	Weak acids	Strong bases	Weak bases
hydrochloric acid (HCl)	ethanoic acid (CH_3COOH)	sodium hydroxide ($NaOH$)	ammonia (NH_3)
sulfuric acid (H_2SO_4)	carbonic acid (H_2CO_3)	potassium hydroxide (KOH)	
nitric acid (HNO_3)	phosphoric acid (H_3PO_4)	calcium hydroxide ($Ca(OH)_2$)	

Strong acids

As you saw previously, when hydrogen chloride gas (HCl) is bubbled through water, it ionises completely—virtually no HCl molecules remain in the solution (Figure 17.2.4a). Similarly, pure HNO_3 which is a covalent molecular compound, also ionises completely in water:

$$HCl(g) \rightarrow H^+(aq) + Cl^-(aq)$$

$$HNO_3(l) \rightarrow H^+(aq) + NO_3^-(aq)$$

The single reaction arrow $\rightarrow$ in each equation above indicates that the ionisation reaction is complete.

Acids that are completely ionised are called **strong acids**. Therefore, solutions of strong acids contain ions, with virtually no unreacted acid molecules remaining. Hydrochloric acid, sulfuric acid and nitric acid are the most common strong acids.

Weak acids

Vinegar is a solution of ethanoic acid. Pure ethanoic acid is a polar covalent molecular compound that ionises in water to produce hydrogen ions and ethanoate (acetate) ions. In a $1.0\,mol\,L^{-1}$ solution of ethanoic acid (CH_3COOH), only a small proportion of ethanoic acid molecules are ionised at any one time (Figure 17.2.4b). A $1.0\,mol\,L^{-1}$ solution of ethanoic acid contains a high proportion of CH_3COOH molecules and only some hydronium ions and ethanoate ions. At 25°C, in a $1.0\,mol\,L^{-1}$ solution of ethanoic acid, the concentrations of $CH_3COO^-(aq)$ and H_3O^+ are only about $0.004\,mol\,L^{-1}$.

The partial ionisation of a **weak acid** is shown in an equation using reversible (double) arrows:

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

Therefore, ethanoic acid is described as a weak acid in water.

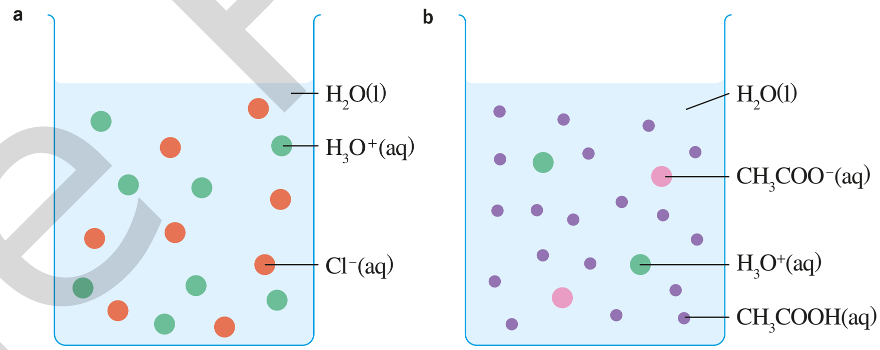


FIGURE 17.2.4 (a) In a $1.0\,mol\,L^{-1}$ solution of a hydrochloric acid the acid molecules are completely ionised in water. (b) However, in a $1.0\,mol\,L^{-1}$ solution of ethanoic acid only a small proportion of the ethanoic acid molecules are ionised.

Strong bases

Sodium hydroxide is a **strong base** as it is completely dissociated into Na^+ and OH^- ions in water. This can be shown by the following equation:

$$NaOH(s) \rightarrow Na^+(aq) + OH^-(aq)$$

Note that as sodium hydroxide is an ionic substance, this process is just described as dissociation. It is not ionisation because the original solid is already comprised of ions which separated from each other when dissolved in water.

Weak bases

Ammonia is a covalent molecular compound that ionises in water to produce hydroxide ions. This ionisation can be represented by the equation:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

Only a small proportion of ammonia molecules are ionised at any instant, so a 1.0 mol L^{-1} solution of ammonia contains mostly ammonia molecules together with a smaller number of ammonium ions and hydroxide ions. This is shown by the double arrow in the equation. Ammonia is a **weak base** in water.

CHEMFILE

Super acids

Fluorosulfuric acid (HSO_3F) is one of the strongest acids known. It has a similar geometry to that of the sulfuric acid molecule (Figure 17.2.5). The highly electronegative fluorine atom causes the oxygen–hydrogen bond in fluorosulfuric acid to be more polarised than the oxygen–hydrogen bond in sulfuric acid. The acidic proton is easily transferred to a base.

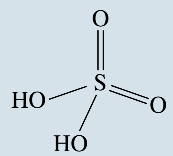


FIGURE 17.2.5 Structure of sulfuric acid (left) and fluorosulfuric acid (right) molecules

Fluorosulfuric acid is classified as a super acid. Super acids are defined as acids that have acidity greater than the acidity of pure sulfuric acid.

Super acids such as fluorosulfuric acid and triflic acid (CF_3SO_3H) are about 1000 times stronger than sulfuric acid. Carborane acid ($H(CHB_{11}Cl_{11})$) is 1 million times stronger than sulfuric acid. The strongest known super acid is fluoroantimonic acid (H_2FSbF_6), which is 10^{16} times stronger than 100% sulfuric acid.

Super acids are used in the production of plastics and high-octane petrol, in coal gasification and in research.

Strength versus concentration

When referring to solutions of acids and bases, it is important not to confuse the terms ‘strong’ and ‘weak’ with ‘concentrated’ and ‘dilute’. Concentrated and dilute describe the amount of acid or base dissolved in a given volume of solution. Hydrochloric acid is a strong acid because the HCl molecules totally ionise in solution. A concentrated solution of hydrochloric acid can be prepared by bubbling a large amount of hydrogen chloride into a given volume of water. By using only a small amount of hydrogen chloride, a dilute solution of hydrochloric acid would be produced.

However, in both cases, the hydrogen chloride is completely ionised—it is a strong acid. Similarly, a solution of ethanoic acid may be concentrated or dilute. However, as it is partially ionised, it is a weak acid (Figure 17.2.6).

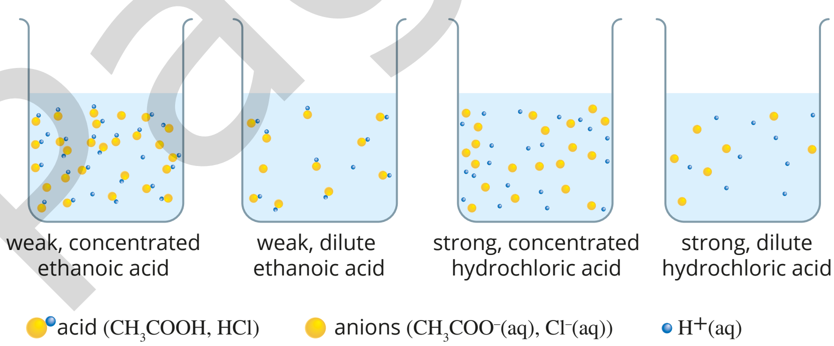


FIGURE 17.2.6 The concentration of ions in an acid solution depends on both the concentration and strength of the acid.

Terms such as ‘weak’ or ‘strong’ acids, or solutions classified as ‘dilute’ or ‘concentrated’ are qualitative (or descriptive) terms. Solutions can be given accurate, quantitative descriptions by stating concentrations in mol L^{-1} or g L^{-1}.

pH AND HYDROGEN ION CONCENTRATION

Definition of pH

The range of H^+ concentrations in solutions is so great that a convenient scale, called the **pH scale**, has been developed to measure acidity. The pH scale was first proposed by the Danish scientist Sören Sörenson in 1909 as a way of expressing levels of acidity. The pH of a solution is defined as:

$$pH = -\log_{10}[H^+]$$

where $[H^+]$ represents the concentration of $H^+(aq)$ ions in the solution, measured in $mol\,L^{-1}$.

Alternatively, this expression can be rearranged to give:

$$[H^+] = 10^{-pH}$$

The pH scale eliminates the need to write cumbersome powers of 10 to describe hydrogen ion concentration. The use of pH greatly simplifies the measurement and calculation of acidity. Since the scale is based upon the negative logarithm of the hydrogen ion concentration, the pH of a solution *decreases* as the concentration of hydrogen ions *increases*.

pH of acidic and basic solutions

Figure 17.2.7 shows a pH meter, which is used to accurately measure the pH of a solution.

Acidic, basic and neutral solutions can be defined in terms of their pH at 25°C.

- **Neutral solutions** have a pH equal to 7.
- **Acidic solutions** have a pH less than 7.
- **Basic solutions** have a pH greater than 7.

On the pH scale, the most acidic solutions have pH values slightly less than 0 and the most basic solutions have values of about 14. The pH values of some common substances are provided in Table 17.2.2.

FIGURE 17.2.7 A pH meter is used to measure the acidity of a solution.

TABLE 17.2.2 pH values of some common substances at 25°C

Solution	pH	$[H_3O^+]$ ($mol\,L^{-1}$)
1.0 $mol\,L^{-1}$ HCl	0.0	1
lemon juice	3.0	10^{-3}
vinegar	4.0	10^{-4}
tomatoes	5.0	10^{-5}
rain water	6.0	10^{-6}
pure water	7.0	10^{-7}
seawater	8.0	10^{-8}
soap	9.0	10^{-9}
oven cleaner	13.0	10^{-13}
1.0 $mol\,L^{-1}$ NaOH	14.0	10^{-14}

A solution with pH 2 has 10 times the concentration of hydronium ions as one of pH 3.

Measurement and understanding of pH are important in a large variety of everyday applications. For example, there is a complex system of pH control in your blood because even small deviations from the normal pH range of 7.35–7.45 for any length of time can lead to serious illness and death.

Calculations involving pH

pH can be calculated using the formula $pH = -\log_{10}[H^+]$. Your scientific calculator has a logarithm function that will simplify pH calculations.

Worked example 17.2.1

CALCULATING THE pH OF AN AQUEOUS SOLUTION FROM $[H^+]$

What is the pH of a solution in which the concentration of H^+ is 0.14 mol L^{-1}? Express your answer to two decimal places.

Thinking	Working
Write down the concentration of $[H^+]$ ions in the solution.	$[H^+] = 0.14$ mol L^{-1}
Substitute the value of $[H^+]$ into: $pH = -\log_{10}[H^+]$ Use the logarithm function on your calculator to determine the answer.	$pH = -\log_{10}[H^+]$ $= -\log_{10}(0.14)$ (use your calculator) $= 0.85$

Worked example: Try yourself 17.2.1

CALCULATING THE pH OF AN AQUEOUS SOLUTION FROM $[H^+]$

What is the pH of a solution in which the concentration of $[H^+]$ is 6×10^{-9} mol L^{-1}?
Express your answer to two significant figures.

You will learn more about pH, including how to calculate the pH of basic solutions in the Year 12 course.

17.2 Review

SUMMARY

- An Arrhenius acid ionises in water to produce hydrogen ions.
- In an ionisation reaction, a molecular substance reacts with a solvent such as water to form ions in solution.
- A molecule of a polyprotic acid can lose more than one hydrogen ion.
- The common properties of acids are due to their ability to form the hydrogen ion, H^+, in solution.
- An Arrhenius base dissociates in water producing hydroxide ions, OH^-.
- In a dissociation process, a molecule or ionic compound separates or splits into smaller particles.
- The common properties of bases are due to the presence of the hydroxide ion, OH^-, in solution.
- Strong acids are completely ionised; weak acids are partially ionised.
- Strong bases are ionic compounds that are completely dissociated. Weak bases are usually molecular compounds that are partially ionised.
- A concentrated acid or base contains more moles of solute per litre than a dilute acid or base.
- The concentration of H^+ is measured using the pH scale:
  $pH = -\log_{10}[H^+]$
- At 25°C the pH of a neutral solution is 7. The pH of an acidic solution is less than 7 and the pH of a basic solution is greater than 7.

KEY QUESTIONS

1. An acid solution is formed when hydrogen bromide gas is bubbled into water. Write a balanced equation for this reaction.
2. Write a balanced equation for the reaction that occurs when H_2SO_4 is completely ionised in water.
3. Which one of the following statements correctly describes an Arrhenius base?
   A a molecular substance that ionises in water, forming H^+ ions
   B a molecular substance that ionises in water, forming OH^- ions
   C an ionic compound that ionises in water, forming OH^- ions
   D an ionic compound that ionises in water, forming H^+ ions
4. Perchloric acid is a stronger acid than ethanoic acid. If you had 1.0 mol L^{-1} solutions of each acid, which acid would you expect to be a better conductor of electricity? Explain why.
5. Calculate the pH of a solution in which $[H^+] = 0.01$ mol L^{-1}.
6. Calculate the pH of a 0.001 mol L^{-1} solution of nitric acid (HNO_3).

17.3 Reactions of acids and bases

Acids were originally grouped together because of their similar chemical behaviour. Chemists use indicators, such as litmus, to identify acidic solutions. Acids and bases react readily with many other chemicals, and some of the early definitions of acids and bases were derived from these reactions.

In this section, you will learn to use the patterns in the reactions of acids and bases to predict the products that are formed.

GENERAL REACTION TYPES INVOLVING ACIDS AND BASES

Acids and bases react in many ways. However, it is possible to group some reactions together according to the similarity of the reactants involved and products formed. While the identification of products should be based on experimental data, these groups, or reaction types, can be useful. The reaction types you will be studying are the reaction of acids with:

- metal hydroxides
- metal carbonates and hydrogen carbonates
- reactive metals

and the reaction of bases with ammonium salts.

ACIDS AND METAL HYDROXIDES

Soluble metal hydroxides, such as NaOH, dissociate in water to form metal cations and hydroxide ions, $OH^-(aq)$. The products of a reaction of an acid with a metal hydroxide are an ionic compound, called a **salt**, and water.

The general rule for the reaction between acids and metal hydroxides can be expressed as:

$$\text{acid} + \text{metal hydroxide} \rightarrow \text{salt} + \text{water}$$

For example, solutions of sulfuric acid and sodium hydroxide react to form sodium sulfate and water. This can be represented by the full (or overall) equation:

$$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

The salt formed in the reaction between sulfuric acid and sodium hydroxide is sodium sulfate. If water was evaporated from the reaction mixture, solid sodium sulfate would be left behind.

Salts

Salts consist of the positive ion or cation from the base and the negative ion or anion from the acid. In Chapter 4, you saw that the names of anions that contain oxygen often end in '-ate'; for example, sulfate, phosphate. The names of anions that do not contain oxygen end in '-ide'; for example, chloride, bromide.

Table 17.3.1 lists the names of salts formed from some neutralisation reactions of acids with metal hydroxides. Note that the name of the positive ion is listed first and the name of the negative ion from the acid is listed second.

TABLE 17.3.1 Salts formed from some common neutralisation reactions

Reactants (acid + metal hydroxide)	Name of salt formed	Formulae of ions present in the salt solution
hydrochloric acid + potassium hydroxide	potassium chloride	$K^+(aq) + Cl^-(aq)$
hydrochloric acid + magnesium hydroxide	magnesium chloride	$Mg^{2+}(aq) + Cl^-(aq)$
nitric acid + sodium hydroxide	sodium nitrate	$Na^+(aq) + NO_3^-(aq)$
sulfuric acid + zinc hydroxide	zinc sulfate	$Zn^{2+}(aq) + SO_4^{2-}(aq)$
phosphoric acid + potassium hydroxide	potassium phosphate	$K^+(aq) + PO_4^{3-}(aq)$
ethanoic acid + calcium hydroxide	calcium ethanoate	$Ca^{2+}(aq) + CH_3COO^-(aq)$

Salts is the general name given to an ionic compound. They are therefore formed from a positive metal cation and a negative non-metal anion. Salts are often formed from reactions involving acids and bases.

IONIC EQUATIONS

The hydroxide ions from metal hydroxides, such as sodium hydroxide (NaOH), calcium hydroxide ($Ca(OH)_2$) and magnesium hydroxide ($Mg(OH)_2$), react readily with the hydrogen ion ($H^+(aq)$) from acids.

The reaction between an acid and a metal hydroxide can be represented by an **ionic equation** as well as by an overall equation.

When writing ionic equations remember that:

- ionic equations are balanced with respect to both the number of atoms of each element and charge
- **spectator ions** are ions that are dissolved in the solution and are present as ions before and after the reaction, but are not involved in the reaction. These are omitted from the ionic equation
- if a reactant or product is a solid, liquid or a gas, it cannot be written as ions and it must be present in the ionic equation.

You also need to remember when writing ionic equations for neutralisation reactions:

- strong acids ionise in solution and are written as ions; for example, HCl in solution is written as $H^+(aq) + Cl^-(aq)$.
- metal hydroxides and salts are ionic and, if soluble, dissociate in solution and are written as ions; for example, KOH dissolving in water is written as:

$$KOH(s) \rightarrow K^+(aq) + OH^-(aq)$$

- water is a covalent molecular substance that does not ionise to any significant extent. It is written as $H_2O(l)$.

Figure 17.3.1 is a diagrammatic representation of the ions in solutions of HCl and NaOH when mixed in a neutralisation reaction.

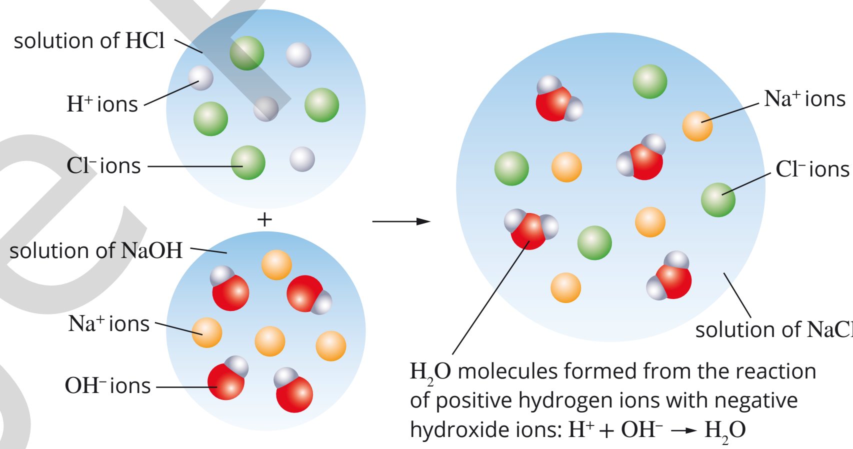


FIGURE 17.3.1 A representation of the reaction between $H^+(aq)$ and $OH^-(aq)$ ions that occurs when solutions of HCl and NaOH are mixed

Worked example 17.3.1 indicates the steps to follow when writing ionic equations for the reactions of acids with bases—**acid**—base reactions.

Worked example 17.3.1

WRITING AN IONIC EQUATION FOR AN ACID–BASE REACTION

Write an ionic equation for the reaction that occurs when hydrochloric acid is added to sodium hydroxide solution. A representation of this reaction is shown in Figure 17.3.1.	
Thinking	**Working**
What is the general reaction? Identify the products formed.	acid + metal hydroxide → salt + water A solution of sodium chloride and water is formed.
Identify the reactants and products. Indicate the state of each, i.e. (aq), (s), (l) or (g).	Reactants: HCl(aq) is ionised in solution, forming $H^+(aq)$ and $Cl^-(aq)$. NaOH(aq) is dissociated in solution, forming $Na^+(aq)$ and $OH^-(aq)$. Products: Sodium chloride is dissociated and exists as $Na^+(aq)$ and $Cl^-(aq)$. Water is a molecular compound and its formula is $H_2O(l)$.
Write the equation showing all reactants and products, in ionised form where possible. (There is no need to balance the equation yet.)	$H^+(aq) + Cl^-(aq) + Na^+(aq) + OH^-(aq) \rightarrow Na^+(aq) + Cl^-(aq) + H_2O(l)$
Identify the spectator ions: the ions that have an (aq) state both as a reactant and as a product.	$Na^+(aq)$ and $Cl^-(aq)$
Rewrite the equation without the spectator ions. Balance the equation with respect to number of atoms of each element and charge.	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

Worked example: Try yourself 17.3.1

WRITING AN IONIC EQUATION FOR AN ACID–BASE REACTION

Write an ionic equation for the reaction that occurs when sulfuric acid is added to potassium hydroxide solution.

When writing ionic equations, if a reactant or product is a solid, liquid or a gas, it cannot be written as ions and it must be present in the ionic equation.

NEUTRALISATION REACTIONS

If a solution of a metal hydroxide is added to a solution of an acid, the hydroxide ions will react with the hydronium ions. The acid and base are said to have been **neutralised** at the point when all the hydroxide ions have reacted with all the hydrogen ions, forming water (H_2O).

CHEMISTRY IN ACTION

Benefits of neutralisation

A neutralisation reaction enables the adjustment of the acidity of a solution. If excess acid is harmful, it can be neutralised by adding a base. Conversely, an environment that is too alkaline can be neutralised by adding an acid. The following are some common examples.

- Methanoic acid (otherwise known as formic acid) is released from the sting of ants, bees and nettles (Figure 17.3.2). If affected skin is rinsed with limewater or a dilute solution of ammonia, the acid becomes neutralised and is no longer painful to the person affected.
- The venom from wasps is alkaline. A common treatment for wasp bites is to rinse the site of the bite with vinegar (ethanoic acid, CH_3COOH) as this neutralises the base within the venom.
- Excess acid in the stomach is the cause of indigestion. This condition is treated with substances that neutralise acids. Antacid tablets are bases that neutralise the excess acid in the stomach.
- The bacteria occurring on tooth enamel feed on the sugars present in food. The products of their metabolism are acids that attack the enamel, which leads to tooth decay. This is why toothpastes (Figure 17.3.3) are weak bases.

FIGURE 17.3.3 A selection of toothpastes. These products are advertised as providing protection against tooth cavities, gum disease and tooth decay.

FIGURE 17.3.2 (a) Irritation on the leg of a person bitten by ants. (b) Scanning electron microscope image of a nettle (*Urtica* sp.) surface showing stinging, hair-like structures (colourised)

ACIDS AND METAL CARBONATES

The weathering of buildings and statues (Figure 17.3.4) is due in part to the reaction between acid rain and the carbonate minerals in the stone.

Acids reacting with metal carbonates and metal hydrogencarbonates produce carbon dioxide gas, together with a salt and water. Metal carbonates include sodium carbonate (Na_2CO_3), magnesium carbonate ($MgCO_3$) and calcium carbonate ($CaCO_3$).

The general reaction for metal carbonates with acids can be summarised as:

$$\text{acid} + \text{metal carbonate} \rightarrow \text{salt} + \text{water} + \text{carbon dioxide}$$

For example, a solution of hydrochloric acid reacting with sodium carbonate solution produces a solution of sodium chloride, water and carbon dioxide gas. The reaction is represented by the equation:

$$2HCl(aq) + Na_2CO_3(aq) \rightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$$

The reaction between hydrochloric acid and sodium carbonate is represented in Figure 17.3.5.

FIGURE 17.3.4 This statue has been heavily eroded by acid rain, which reacts with carbonate salts in limestone. Acid rain is formed when gases, such as sulfur dioxide and nitrogen oxides, dissolve in water to form acidic solutions.

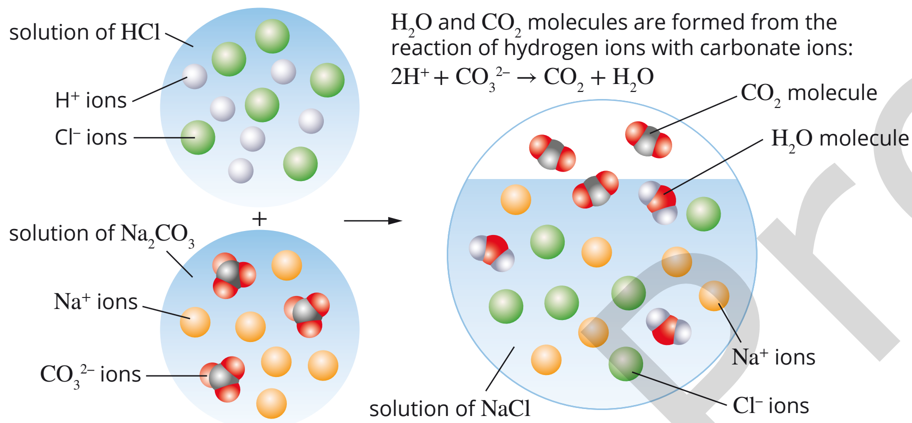


FIGURE 17.3.5 The reaction between solutions of sodium carbonate and hydrochloric acid

Metal hydrogencarbonates (also known as bicarbonates) include sodium hydrogencarbonate ($NaHCO_3$), potassium hydrogencarbonate ($KHCO_3$) and calcium hydrogencarbonate ($Ca(HCO_3)_2$). Acids added to metal hydrogencarbonates also produce carbon dioxide together with a salt and water. The general reaction is:

$$\text{acid} + \text{metal hydrogencarbonate} \rightarrow \text{salt} + \text{water} + \text{carbon dioxide}$$

For example, when a solution of hydrochloric acid and solid sodium hydrogencarbonate are mixed, the following reaction occurs:

$$HCl(aq) + NaHCO_3(s) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)$$

The reactions between acids and metal carbonates, and reactions between acids and metal hydrogencarbonates, can also be represented as ionic equations by following steps similar to the steps for writing reactions between acids and bases.

CHEMFILE

Bicarbonate of soda

Self-raising flour contains tartaric acid and some sodium hydrogencarbonate (bicarbonate of soda) (Figure 17.3.6). It is used in baking cakes because on heating in the oven, the acid and hydrogencarbonate react. Carbon dioxide is released, which causes the cake mixture to rise.

FIGURE 17.3.6 Bicarbonate of soda leads to the production of carbon dioxide causing these muffins to rise.

Worked example 17.3.2

WRITING IONIC EQUATIONS FOR REACTIONS BETWEEN ACIDS AND METAL CARBONATES

What products are formed when a dilute solution of nitric acid is added to solid magnesium carbonate? Write an ionic equation for this reaction.	
Thinking	**Working**
What is the general reaction? Identify the products.	acid + metal carbonate → salt + water + carbon dioxide Products of this reaction are magnesium nitrate in solution, water and carbon dioxide gas.
Identify the reactants and products. Indicate the state of each, i.e. (aq), (s), (l) or (g).	Reactants: Nitric acid is ionised in solution, forming $H^+(aq)$ and $NO_3^-(aq)$ ions. Magnesium carbonate is an ionic solid, $MgCO_3(s)$. Products: Magnesium nitrate is dissociated into $Mg^{2+}(aq)$ and $NO_3^-(aq)$ ions. Water has the formula $H_2O(l)$. Carbon dioxide has the formula $CO_2(g)$.
Write the equation showing all reactants and products. (There is no need to balance the equation yet.)	$H^+(aq) + NO_3^-(aq) + MgCO_3(s) \rightarrow Mg^{2+}(aq) + NO_3^-(aq) + H_2O(l) + CO_2(g)$
Identify the spectator ions.	$NO_3^-(aq)$
Rewrite the equation without the spectator ions. Balance the equation with respect to number of atoms of each element and charge.	The equation with spectator ions omitted is: $H^+(aq) + MgCO_3(s) \rightarrow Mg^{2+}(aq) + H_2O(l) + CO_2(g)$ The balanced equation is: $2H^+(aq) + MgCO_3(s) \rightarrow Mg^{2+}(aq) + H_2O^+(l) + CO_2(g)$

Worked example: Try yourself 17.3.2

WRITING IONIC EQUATIONS FOR REACTIONS BETWEEN ACIDS AND METAL CARBONATES

What products are formed when a solution of hydrochloric acid is added to a solution of sodium hydrogencarbonate? Write an ionic equation for this reaction.

FIGURE 17.3.7 Limewater test. Carbon dioxide is bubbled through limewater, turning the limewater cloudy.

TESTING FOR CARBONATE SALTS

Acids can be used to detect the presence of carbonate salts. Carbon dioxide is produced when an acid is added to a carbonate.

The **limewater test** is a simple laboratory test used to confirm the presence of carbon dioxide gas. Limewater is a saturated solution of calcium hydroxide ($Ca(OH)_2$). When carbon dioxide is bubbled through this solution, it will turn 'milky' or 'cloudy' in appearance (Figure 17.3.7) due to the precipitation of calcium carbonate:

$$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$$

ACIDS AND REACTIVE METALS

When dilute acids are added to main group metals and some transition metals, bubbles of hydrogen gas are released and a salt is formed. In general, the equation for the reaction is:

$$\text{acid} + \text{reactive metal} \rightarrow \text{salt} + \text{hydrogen}$$

Reactive metals include calcium, magnesium, iron and zinc, but *not* copper, silver or gold. For example, the reaction between dilute hydrochloric acid and zinc metal can be seen in Figure 17.3.8 and represented by the equation:

$$2HCl(aq) + Zn(s) \rightarrow ZnCl_2(aq) + H_2(g)$$

This reaction can also be represented by an ionic equation. In an aqueous solution, the hydrochloric acid is ionised and the salt, zinc chloride (a soluble ionic compound), is dissociated. The ionic equation can be determined as shown in Worked example 17.3.3.

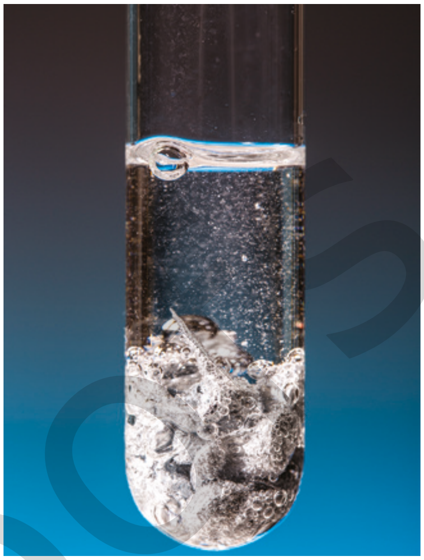


FIGURE 17.3.8 Hydrogen gas is produced from the reaction of zinc with dilute hydrochloric acid.

Worked example 17.3.3

WRITING IONIC EQUATIONS FOR REACTIONS BETWEEN ACIDS AND REACTIVE METALS

Write an ionic equation for the reaction that occurs when dilute hydrochloric acid is added to a sample of zinc metal.	
Thinking	**Working**
What is the general reaction? Identify the products formed.	acid + reactive metal → salt + hydrogen Hydrogen gas and zinc chloride solution are produced.
Identify the reactants and products. Indicate the state of each, i.e. (aq), (s), (l) or (g).	Reactants: zinc is a solid, Zn(s). Hydrochloric acid is ionised, forming $H^+(aq)$ and $Cl^-(aq)$ ions. Products: hydrogen gas, $H_2(g)$. Zinc chloride is dissociated into $Zn^{2+}(aq)$ and $Cl^-(aq)$ ions.
Write the equation showing all reactants and products. (There is no need to balance the equation yet.)	$H^+(aq) + Cl^-(aq) + Zn(s) \rightarrow Zn^{2+}(aq) + Cl^-(aq) + H_2(g)$
Identify the spectator ions.	$Cl^-(aq)$
Rewrite the equation without the spectator ions. Balance the equation with respect to number of atoms of each element and charge.	$2H^+(aq) + Zn(s) \rightarrow Zn^{2+}(aq) + H_2(g)$

Worked example: Try yourself 17.3.3

WRITING IONIC EQUATIONS FOR REACTIONS BETWEEN ACIDS AND REACTIVE METALS

Write an ionic equation for the reaction that occurs when aluminium is added to a dilute solution of hydrochloric acid.

BASES WITH AMMONIUM SALTS

Ammonium salts are similar to metal salts except that the metal cation is replaced by the ammonium ion (NH_4^+). Some examples of ammonium salts are ammonium chloride (NH_4Cl) ammonium nitrate (NH_4NO_3) and ammonium sulfate ($(NH_4)_2SO_4$). Ammonium salts are highly soluble in water and dissociate into their constituent ions.

Ammonia gas (NH_3) and water are formed when bases react with ammonium salts:

$$\text{ammonium salt} + \text{base} \rightarrow \text{salt} + \text{ammonia} + \text{water}$$

For example, when a solution of sodium hydroxide is added to a solution of ammonium sulfate, sodium sulfate, ammonia gas and water are produced. The ammonia can be detected by its characteristic odour or by reaction with moist red litmus paper, which turns blue in the presence of ammonia.

The equation for this reaction is:

$$2NaOH(aq) + (NH_4)_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2NH_3(g) + 2H_2O(l)$$

The ionic equation can be written using the steps outlined earlier in this chapter. Remember that NaOH, $(NH_4)_2SO_4$ and Na_2SO_4 are ionic solids that dissociate in water and that NH_3 and H_2O are molecular compounds. Na^+ and SO_4^{2-} are spectator ions and take no part in this reaction. The ionic equation simplifies to:

$$NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(g) + H_2O(l)$$

17.3 Review

SUMMARY

- Generalisations can be made about the likely products of reactions involving acids and bases:
  - acid + metal hydroxide → salt + water
  - acid + metal carbonate → salt + water + carbon dioxide
  - acid + metal hydrogencarbonate → salt + water + carbon dioxide
  - acid + reactive metal → salt + hydrogen
  - base + ammonium salt → salt + ammonia gas + water
- Each of these reactions can be represented by full and ionic equations.
- An ionic equation only shows those ions, atoms or molecules that take part in the reaction.
- Spectator ions (ions that do not take part in the reaction) are not included in ionic equations.
- Ionic equations are balanced with respect to both the number of atoms of each element and charge.

KEY QUESTIONS

1 Write full and ionic chemical equations for the reactions between:

a magnesium and sulfuric acid
b calcium and hydrochloric acid
c zinc and ethanoic acid
d aluminium and nitric acid.

2 Name the salt produced in each of the reactions in Question **1**.

3 For each of the following reactions, write:

i a full chemical equation to represent the reaction (remember to include states)
ii an ionic equation.

a solid zinc oxide and sulfuric acid
b solid calcium and nitric acid
c solid copper(II) hydroxide and nitric acid
d solid magnesium hydrogencarbonate and hydrochloric acid
e solid tin(II) carbonate and sulfuric acid

4 Predict the products of the following reactions and write full and ionic chemical equations for each.

a A solution of sulfuric acid is added to a solution of potassium hydroxide.
b Nitric acid solution is mixed with sodium hydroxide solution.
c Hydrochloric acid solution is poured onto some solid magnesium hydroxide.
d Blue copper(II) carbonate powder is added to dilute sulfuric acid.
e Dilute hydrochloric acid is mixed with a solution of potassium hydrogencarbonate.
f Dilute nitric acid is added to a spoon coated with solid zinc.
g Hydrochloric acid solution is added to some marble chips (calcium carbonate).
h Solid bicarbonate of soda (sodium hydrogencarbonate) is mixed with vinegar (a dilute solution of ethanoic acid).
i A solution of ammonium carbonate is added to a solution of calcium hydroxide. The mixture is then warmed.

17.4 Calculations involving acids and bases

Although acids are frequently purchased as concentrated solutions, you will often need to use them in a more diluted form. For example, a bricklayer uses a 10% solution of hydrochloric acid to remove mortar splashes from bricks used to build a house. The brick-cleaning solution is prepared by diluting concentrated hydrochloric acid by a factor of ten.

In this section, you will learn how to calculate the concentration and pH of acids and bases once they have been diluted.

CONCENTRATION OF ACIDS AND BASES

The concentration of acids and bases is usually expressed in units of $mol\,L^{-1}$ (or M). This is also referred to as molar concentration or molarity.

You will recall from Chapter 16 that the molar concentration of a solution, in $mol\,L^{-1}$, is given by the expression:

$$c = \frac{n}{V}$$

where c is the molar concentration ($mol\,L^{-1}$), n is the amount of solute (mol) and V is the volume of the solution (L).

The most convenient way of preparing a solution of a dilute acid is by mixing concentrated acid with water, as shown in Figure 17.4.1. This is known as a **dilution**.

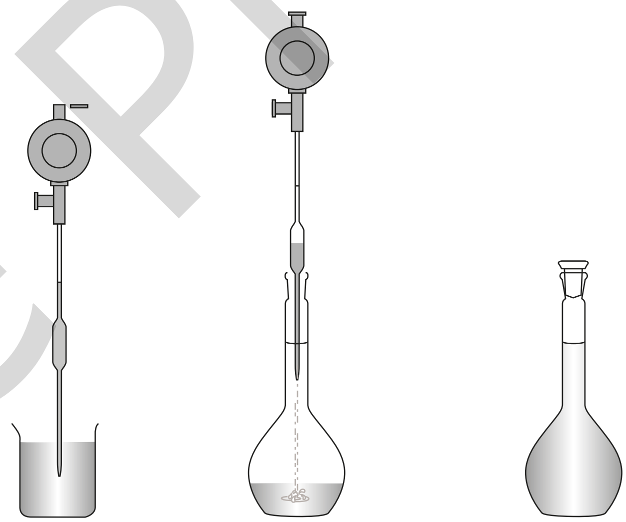


FIGURE 17.4.1 Preparing a $1.00\,mol\,L^{-1}$ HCl solution by diluting a $10.0\,mol\,L^{-1}$ solution. Heat is released when a concentrated acid is added to water, so the volumetric flask is partly filled with water before the acid is added. (Extra safety precautions would be required for diluting concentrated sulfuric acid.)

The amount of solute (in moles) in a solution does not change when a solution is diluted; the volume of the solution increases and the concentration decreases. The change in concentration or volume can be calculated using the formula:

$$c_1V_1 = c_2V_2$$

where c_1 and V_1 are the initial concentration and volume, and c_2 and V_2 are the concentration and volume after dilution.

The concentration of a dilute acid can be calculated if the following are known:

- volume of the concentrated solution
- concentration (molarity) of the concentrated solution
- total volume of water added.

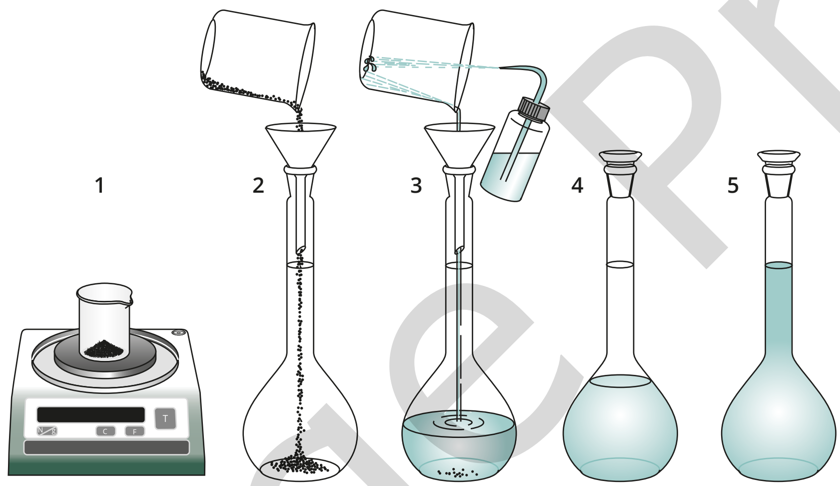
The molar concentration of some concentrated acids are shown in Table 17.4.1.

TABLE 17.4.1 Molar concentrations of some concentrated acids

Concentrated acid (% by mass)	Formula	Molar concentrations (mol L^{-1})
ethanoic acid (99.5%)	CH_3COOH	17
hydrochloric acid (36%)	HCl	12
nitric acid (70%)	HNO_3	16
phosphoric acid (85%)	H_3PO_4	15
sulfuric acid (98%)	H_2SO_4	18

In the laboratory, you can prepare solutions of a base of a required concentration by:

- diluting a more concentrated solution, or
- dissolving a weighed amount of the base in a measured volume of water, as shown in Figure 17.4.2.



1 Accurately weigh out a mass of the base.
2 Transfer the base to a volumetric flask.
3 Ensure complete transfer of the base by washing with water.
4 Dissolve the base in water.
5 Add water to make the solution up to the calibration mark and shake thoroughly.

FIGURE 17.4.2 Preparing a solution by dissolving a weighed amount of base in a measured volume of water

Worked example 17.4.1

CALCULATING MOLAR CONCENTRATION AFTER DILUTION

Calculate the molar concentration of hydrochloric acid when 10.0 mL of water is added to 5.0 mL of 1.2 mol L^{-1} HCl.	
Thinking	**Working**
The number of moles of solute does not change during a dilution. So $c_1V_1 = c_2V_2$, where c is the concentration in mol L^{-1} and V is the volume of the solution. (Each of the volume units must be the same, although not necessarily litres.)	$c_1V_1 = c_2V_2$
Identify given values for concentrations and volumes before and after dilution. Identify the unknown.	10.0 mL was added to 5.0 mL, the final volume is 15.0 mL. (In practice, small volume changes can occur when solutions are mixed; assume no volume changes occur.) $c_1 = 1.2$ mol L^{-1} $V_1 = 5.0$ mL $V_2 = 15.0$ mL You are required to calculate, c_2, the concentration after dilution.
Transpose the equation and substitute the known values into the equation to find the required value.	$c_2 = \frac{c_1 \times V_1}{V_2}$ $= 1.2 \times \frac{5.0}{15.0}$ $= 0.40$ mol L^{-1}

Worked example: Try yourself 17.4.1

CALCULATING MOLAR CONCENTRATION AFTER DILUTION

Calculate the molar concentration of nitric acid when 80.0 mL of water is added to 20.0 mL of 5.00 mol L^{-1} HNO_3.

Worked example 17.4.2

CALCULATING THE VOLUME OF WATER TO BE ADDED IN A DILUTION

How much water must be added to 30.0 mL of 2.50 mol L^{-1} HCl to dilute the solution to 1.00 mol L^{-1}?	
Thinking	**Working**
The number of moles of solute does not change during a dilution. So $c_1V_1 = c_2V_2$, where c is the concentration in mol L^{-1} and V is the volume of the solution. (Each of the volume units must be the same, although not necessarily litres.)	$c_1V_1 = c_2V_2$
Identify given values for concentrations and volumes before and after dilution. Identify the unknown.	$c_1 = 2.50$ mol L^{-1} $V_1 = 30.0$ mL $c_2 = 1.00$ mol L^{-1} You are required to calculate V_2, the volume of the diluted solution.

Transpose the equation and substitute the known values into the equation to find the required value.	$V_2 = \frac{c_1 \times V_1}{c_2}$ $= \frac{2.50 \times 30.0}{1.00}$ $= 75.0\,\text{mL}$
Calculate the volume of water to be added.	Volume of dilute solution = 75.0 mL Initial volume of acid = 30.0 mL So 75.0 – 30.0 = 45.0 mL of water must be added.

Worked example: Try yourself 17.4.2

CALCULATING THE VOLUME OF WATER TO BE ADDED IN A DILUTION

How much water must be added to 15.0 mL of 10.0 mol L^{-1} NaOH to dilute the solution to 2.00 mol L^{-1}?

EFFECT OF DILUTION ON pH OF STRONG ACIDS AND BASES

Consider a 0.1 mol L^{-1} solution of hydrochloric acid. HCl is a strong acid, so the concentration of H^+ ions in this solution is 0.1 mol L^{-1}. Since $pH = -\log_{10}[H^+]$, the pH of this solution is 1.0 (see Section 17.2 if you need to review this formula).

If a 1.0 mL solution is diluted by a factor of 10 to 10.0 mL by the addition of 9.0 mL of water, the concentration of H^+ ions decreases to 0.01 mol L^{-1} and the pH increases to 2.0. A further dilution by a factor of 10 to 100 mL will increase pH to 3.0. However, note that when acids are repeatedly diluted, the pH cannot increase above 7.

We will now look at how to calculate the pH of solutions of strong acids after dilution. (A discussion of the effect of dilution on the pH of solutions of weak acids and bases is complex and beyond the scope of this course.)

Worked example 17.4.3

CALCULATING THE pH OF A DILUTED ACID

5.0 mL of 0.010 mol L^{-1} HNO_3 is diluted to 100.0 mL. What is the pH of the diluted solution?

Thinking	Working
Identify given values for concentrations and volumes before and after dilution.	$c_1 = 0.010\,\text{mol L}^{-1}$ $V_1 = 5.0\,\text{mL}$ $V_2 = 100.0\,\text{mL}$ $c_2 = ?$
Calculate c_2, which is the concentration of H^+ after dilution, by transposing the formula: $c_1V_1 = c_2V_2$	$c_2 = \frac{c_1 \times V_1}{V_2}$ $= \frac{0.010 \times 5.0}{100.0}$ $= 0.00050\,\text{mol L}^{-1}$
Calculate pH using: $pH = -\log_{10}[H^+]$ Use the logarithm function on your calculator to determine pH.	$pH = -\log_{10}[H^+]$ $= -\log_{10}(0.00050)$ $= 3.30$

Worked example: Try yourself 17.4.3

CALCULATING THE pH OF A DILUTED ACID

10.0 mL of 0.1 mol L^{-1} HCl is diluted to 30.0 mL. Calculate the pH of the diluted solution.

REACTING QUANTITIES OF ACIDS AND BASES

Calculations based on the reactions of acids usually involve determining the number of mole of a substance.

You will recall the following from earlier chapters.

- The coefficients in a balanced chemical equation gives the ratio in which substances react.
- The amount of solid, liquid or gas (in mole) can be calculated from the expression $n = \frac{m}{M}$, where m is the mass in gram and M is the molar mass in g mol^{-1}.
- The amount of solute in a solution is given by $n = cV$, where c is the concentration in mol L^{-1} and V is the volume in litres.
- The amount, in mol, of a gas at STP is given by $n = \frac{V}{22.71}$, where V is the volume of the gas in litres at STP.

The major steps in solving calculation problems involving acids are bases are as follows:

1. Write a balanced equation for the reaction.
2. Calculate the amount, in mol, of the substance with the known volume and concentration.
3. Use the mole ratio from the equation to calculate the amount, in mol, of the required substance.

The steps in a stoichiometric calculation are summarised in Figure 17.4.3.

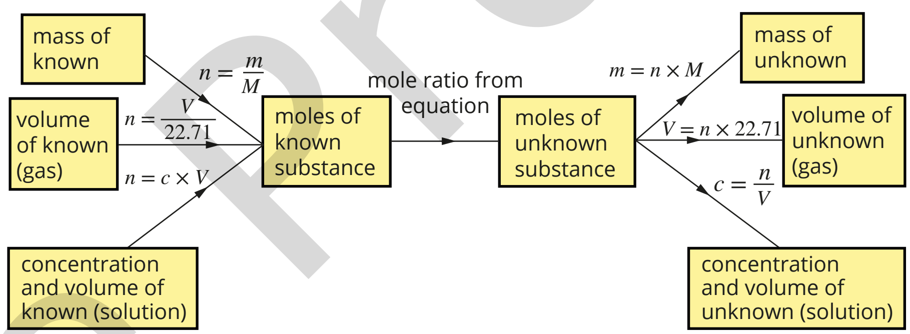


FIGURE 17.4.3 Flow chart for mass, solution and gas calculations involving reactions

Worked example 17.4.4

DETERMINING THE VOLUME OF AN UNKNOWN SOLUTION FROM A NEUTRALISATION REACTION

What volume of 0.100 mol L^{-1} sulfuric acid reacts completely with 178 mL of 0.150 mol L^{-1} potassium hydroxide?	
Thinking	**Working**
Write a balanced chemical equation.	$2KOH(aq) + H_2SO_4(aq) \rightarrow K_2SO_4(aq) + 2H_2O(l)$
Calculate the amount, in mol, of the solution of known concentration and volume.	$n(KOH) = cV = 0.150 \times 0.0178 = 0.00267$ mol
Determine the mole ratio as given by the balanced chemical equation.	Mole ratio $= \frac{n(\text{unknown})}{n(\text{known})} = \frac{1}{2}$
Determine the amount, in mol, of the unknown solution.	$n(H_2SO_4) = 0.00267 \times \frac{1}{2} = 0.00134$ mol
Determine the volume of the unknown solution needed to react.	$n(H_2SO_4) = \frac{n}{V} = \frac{0.00134}{0.100} = 0.0134\ L = 13.4\ mL$ Therefore, 13.4 mL of H_2SO_4 is required to react with 178 mL of 0.150 mol L^{-1} KOH.

Worked example: Try yourself 17.4.4

DETERMINING THE VOLUME OF AN UNKNOWN SOLUTION FROM A NEUTRALISATION REACTION

Calculate the volume of 0.500 mol L^{-1} hydrochloric acid (HCl), that reacts completely with 25.0 mL of 0.100 mol L^{-1} calcium hydroxide ($Ca(OH)_2$) solution.

STOICHIOMETRY PROBLEMS INVOLVING EXCESS REACTANTS

Reactants are not always mixed in stoichiometric amounts in acid–base reactions. There are occasions when one of the reactants, an acid or a base, is in excess. To determine the **limiting reactant** and the amount of the unknown substance:

- calculate the number of moles of each reactant
- determine which reactant is in excess and which is the limiting reactant
- use the amount of limiting reactant to work out the amount of product formed or the amount of reactant in excess.

Worked example 17.4.5

A LIMITING REACTANT ACID–BASE PROBLEM

20.0 mL of a 1.00 mol L^{-1} LiOH solution is added to 30.0 mL of a 0.500 mol L^{-1} HNO_3 solution.

a Which reactant is the limiting reactant?

b What mass of $LiNO_3$ will the reaction mixture contain when the reaction is complete?

Thinking	Working
Write a balanced chemical equation.	$LiOH(aq) + HNO_3(aq) \rightarrow LiNO_3(aq) + H_2O(l)$
Calculate the amount, in mol, of the first reactant.	$n(LiOH) = cV = 0.100 \times 0.0200 = 0.00200$ mol
Calculate the amount, in mol, of the second reactant.	$n(HNO_3) = cV = 0.500 \times 0.0300 = 0.0150$ mol
Use the coefficients of the reaction to determine the limiting reactant.	The equation shows that 1 mole of HNO_3 reacts with 1 mole of LiOH. So the HNO_3 is the limiting reactant.
Find the mole ratio of the unknown substance to the limiting reactant from the equation coefficients.	mole ratio $= \frac{n(\text{unknown})}{n(\text{known})} = \frac{1}{1}$
Determine the amount, in mol, of the unknown substance.	$n(LiNO_3) = 0.00150$ mol
Determine the mass, in g, of the unknown substance.	$(LiNO_3) = nM = 0.00150 \times 68.98 = 1.035$ g $= 1.04$ g

Worked example: Try yourself 17.4.5

A LIMITING REACTANT ACID-BASE PROBLEM

30.0 mL of a 0.100 mol L^{-1} H_2SO_4 solution is mixed with 40.0 mL of a 0.200 mol L^{-1} KOH solution.

a Which reactant is the limiting reactant?

b What will be the mass of K_2SO_4 produced by this reaction?

17.4 Review

SUMMARY

- Amounts of acid or base in solution do not change during a dilution. The volume of the solution increases and its concentration decreases.
- Solutions of acids and bases of a required concentration can be prepared by diluting more concentrated solutions using the formula:

$$c_1V_1 = c_2V_2$$

where c_1 and V_1 are the initial concentration and volume, and c_2 and V_2 are the final concentration and volume after dilution.
- The pH increases when a solution of an acid is diluted.
- The pH of a diluted acid can be determined by calculating the concentration of hydrogen ions in the diluted solution.
- The pH decreases when a solution of a base is diluted.
- Given the quantity of one of the reactants or products in a chemical reaction, the quantity of all other reactants and products can be predicted by working through the following steps.
  1. Write a balanced equation for the reaction.
  2. Calculate the amount (in mol) of the given substance.
  3. Use the mole ratios of reactants and products in the balanced chemical equation to calculate the amount (in mol) of the required substance.
  4. Convert the amount (in mol) of the required substance to the units required in the question.

KEY QUESTIONS

1 1.0 L of water is added to 3.0 L of 0.10 mol L^{-1} HCl. What is the concentration of the diluted acid?

2 How much water must be added to 10 mL of a 2.0 mol L^{-1} sulfuric acid solution to dilute it to 0.50 mol L^{-1}?

3 What volume of water must be added to dilute a 20.0 mL volume of 0.600 mol L^{-1} HCl to 0.100 mol L^{-1}?

4 Describe the effect on the pH of a monoprotic acid solution of pH 1.0 when it is diluted by a factor of 10.

5 10.0 mL of a 0.200 mol L^{-1} sulfuric acid solution is added to 16.0 mL of a 0.100 mol L^{-1} sodium carbonate solution. The equation for the reaction is:

$$Na_2CO_3(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + CO_2(g) + H_2O(l)$$

a Calculate the number of moles of H_2SO_4 in the sulfuric solution.

b Calculate the number of moles of Na_2CO_3 in the sodium carbonate solution.

c Identify the limiting reactant and excess reactant.

d Calculate, the amount in mol, of excess reactant that remains unreacted.

6 18.26 mL of dilute nitric acid reacts completely with 20.00 mL of 0.09927 mol L^{-1} potassium hydroxide solution.

a Write a balanced chemical equation for the reaction between nitric acid and potassium hydroxide.

b Calculate the amount, in mol, of potassium hydroxide consumed in this reaction.

c What amount, in mol, of nitric acid reacted with the potassium hydroxide in this reaction?

d Calculate the concentration of the nitric acid.

Chapter review

17

KEY TERMS

acid
acid–base reaction
acidic solution
acidity
alkali
Arrhenius model
base
basic solution
diprotic acid
hydroxide ion
indicator
limewater test
limiting reactant
monoprotic acid
neutral solution
neutralisation reaction
neutralise
pH scale
polyprotic acid
salt
spectator ions
strong acid
strong base
triprotic acid
weak acid
weak base

Properties of acids and bases

1 Describe how you could use red litmus paper to distinguish between an acidic and an alkaline solution.

2 Using suitable examples, distinguish between:

a a diprotic and a monoprotic substance

b a strong acid and a concentrated acid.

3 Draw a structural formula of the monoprotic ethanoic acid molecule. Identify which proton is donated in an acid–base reaction.

The Arrhenius model of acids and bases

4 Describe what is meant by a scientific ‘model’.

5 Using the Arrhenius model of acids and bases, explain the difference between strong and weak acids.

6 Write an equation to show perchloric acid ($HClO_4$) acting as a strong acid.

7 Write an equation to show hypochlorous acid ($HClO_3$) acting as a weak acid.

8 Write an equation to show ammonia (NH_3) acting as a weak base in water.

9 Human blood has a pH of 7.4. Is blood acidic, basic or neutral?

10 The pH of a cola drink is 3 and of black coffee is 5. How many more times acidic is the cola than black coffee?

11 Calculate the pH of each of the following solutions at 25°C.

a $0.01\ mol\ L^{-1}$ solution of nitric acid (HNO_3)

b $0.20\ mol\ L^{-1}$ solution of hydrochloric acid (HCl)

c $2.0\ mol\ L^{-1}$ solution of sulfuric acid (H_2SO_4)

Reactions of acids and bases

12 Complete, and balance, the following chemical equations.

a $HNO_3(aq) + KOH(aq) \rightarrow$

b $H_2SO_4(aq) + K_2CO_3(aq) \rightarrow$

c $H_3PO_4(aq) + Ca(HCO_3)_2(s) \rightarrow$

d $HF(aq) + Zn(OH)_2(s) \rightarrow$

13 Which one of the following correctly identifies all the products formed when magnesium hydroxide reacts with hydrochloric acid?

A water

B chloride ions

C magnesium ions

D magnesium chloride precipitate

E water, magnesium ions and chloride ions

14 Acids can be represented by a general formula HA. Write a balanced ionic equation for the reaction between an acid HA and a soluble metal carbonate MCO_3.

15 Dilute hydrochloric acid is added to a white solid in a test-tube. A colourless gas is produced. The gas turns limewater cloudy. What is a possible identity of the white solid?

16 Hydrogen is produced when dilute sulfuric acid reacts with aluminium metal.

a Write a balanced full equation for this reaction.

b Write a balanced ionic equation for this reaction.

Calculations involving acids and bases

17 A solution of hydrochloric acid has a pH of 2.

a What is the molar concentration of hydrogen ions in the solution?

b What amount of hydrogen ion, in mol, would be present in 500 mL of this solution?

18 Calculate the pH of each of the following mixtures at 25°C.

a 10 mL of $0.025\ mol\ L^{-1}$ HCl is diluted to 50 mL of solution.

b 10 mL of $0.15\ mol\ L^{-1}$ HCl is diluted to 1.5 L of solution.

19 The molarity of concentrated sulfuric acid is $18.0\ mol\ L^{-1}$. What volume of concentrated sulfuric acid is required to prepare a 1.00 L solution of $2.00\ mol\ L^{-1}$ H_2SO_4?

20 40.0 mL of $0.10\ mol\ L^{-1}$ HNO_3 is diluted to 500.0 mL. Will the pH increase or decrease?

Connecting the main ideas

21 A laboratory assistant forgot to label 0.1 mol L^{-1} solutions of sodium hydroxide (NaOH), hydrochloric acid (HCl), glucose ($C_6H_{12}O_6$), ammonia (NH_3) and ethanoic acid (CH_3COOH). In order to identify them, temporary labels A–E were placed on the bottles and the electrical conductivity and pH of each solution was measured. The results are shown in the table below. Identify each solution and briefly explain your reasoning.

Solution	Electrical conductivity	pH
A	poor	11
B	zero	7
C	good	13
D	good	1
E	poor	3

22 Write concise definitions for:

a an Arrhenius base
b strong acid
c molarity.